	Attack Success Rate (*ASR*) (%)				
Methods	GPT-3.5	Claude-3	LLaMA-2-7B	Vicuna-7B	Mistral-7B
"Let's think step by step"	0.0	0.0	0.0	0.0	8.9
Zero-Shot-CoT	10.1	0.0	1.2	22.8	11.4
Manual-CoT	0.0	0.0	0.0	8.9	12.7
Auto-CoT	15.1	0.0	2.5	29.1	18.9
GCG (Transfer)	**40.5**	2.5	39.2	79.7	73.4
CoT-GCG (This paper)	**40.5**	**10.1**	**50.4**	**97.5**	**83.6**

Table 2: Attack success rate (ASR) measured on GPT-3.5 Turbo, Claude-3 Haiku, LLaMA2-7B, Vicuna-7B, and Mistral-7B using Zero-Shot only, Zero-Shot-CoT, Manual-CoT, Auto-CoT, GCG prompt as a suffix, and CoT-GCG prompt as a suffix. The results averaged over 79 high-risk harmful behaviors.

tected against this phenomenon. However, it is relatively weak against "Weapons" and "Regulated" topics, with an average of over 100%. This suggests that the filtering mechanisms of these models were adapted accordingly during the alignment phase, resulting in biases that enhance or diminish the precautionary nature of some topics, which would be exploited by miscreants. Therefore, it is imperative to implement more comprehensive manual interventions and precautions for sensitive topics in specific LLMs in order to minimize or avoid bias against these topics.

3 Conclusion

This work presents the CoT-GCG, a novel approach for crafting universal adversarial attacks on large language models by leveraging the chain of thought reasoning. This method integrates gradient-based adversarial attacks with CoT techniques to enhance attack transferability and effectiveness. Experiments evaluated the performance of each attack method against various publicly aligned LLMs. The experimental results demonstrate that CoT-GCG significantly outperforms non-gradient-based CoT methods and achieves competitive performance compared to the original GCG transfer approach, even under constrained computational resources. Notably, the categorization of harmful outputs by *Llama Guard* exhibits that LLMs become particularly vulnerable to specific harmful topics during adversarial attacks. These findings highlight potential weaknesses in the alignment process and underscore the critical need for ongoing research on LLM safety and the development of more robust alignment strategies to mitigate the biases and risks associated with adversarial attacks.

4 Limitations

The study yielded promising results, but it is important to note that it faced three main limitations that may have influenced the outcomes.

Computational Resource Constraints
The computational resources of 46 GiB slow GPUs were insufficient to train the full parameter set of the original GCG experiment, resulting in inferior outcomes to those reported in the original GCG paper. Furthermore, limited resources prevented testing larger models and various model combinations, potentially hindering more effective attacks on aligned LLMs. Despite these constraints, our enhanced method still outperformed GCG and non-gradient-based CoT algorithms, demonstrating its competitiveness.

Slow Convergence with Longer Prompts
The convergence speed of the algorithm slowed as the length of the prompts increased. This is because the GCG algorithm calculates the gradient for each candidate and updates the gradient of the optimum each time. As the token length increases, the computational load and processing time increase correspondingly. In particular, convergence becomes slower as the algorithm evolves. Given the limitations of computational resources, it is not advisable to trade off performance by consuming an excessive amount of time, which also impedes the validation of various strategies.

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